WIMNet: A Whole-Image-Based Mamba Network for Joint Hyperspectral and LiDAR Classification

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***Abstract*—Although recent advances have improved joint hyperspectral images (HSIs) and light detection and ranging (LiDAR) data classification, most existing approaches adopt a patch-based processing strategy that leads to substantial computational redundancy for high-resolution data. Moreover, existing architectures usually process spectral and spatial features independently, making it difficult to capture their relationships and thus limiting the fusion performance. To address the above challenges, we propose a whole-image-based Mamba network (WIMNet) that performs image-level fusion for joint hyperspectral and LiDAR classification. Unlike existing patch-based approaches, the proposed WIMNet consists of two dedicated branches: a joint spectral–spatial Mamba branch (JSMB), incorporating a spectral-spatial mamba block (SSMB) with a spectral modulation mechanism (SMM) for adaptive spectral–spatial joint learning, and the multiscale LiDAR Mamba branch (MLMB) for LiDAR feature extraction. Then their features are integrated via a multimodal feature fusion module (MFFM) to produce the final classification map. Final extensive experiments on Houston2013, MUUFL, and Trento datasets demonstrate that WIMNet achieves state-of-the-art accuracy while reducing GFLOPs and inference time by up to 99% compared with existing 11 leading methods, confirming its remarkable efficiency in joint hyperspectral and LiDAR classification. Code will be available at https://github.com/Dufu9/WIMNet**

***Index Terms*—Hyperspectral images (HSIs), light detection and ranging (LiDAR), Mamba, spatial–spectral, whole-image.**

I. Introduction

In RECENT years, remote sensing (RS) technology has developed rapidly, which provides researchers with a large volume of multisource and multimodal RS data [1]. The increasing availability of such data enables the characterization of surface features with higher precision and richer dimensionality. Hyperspectral image (HSI) [2], [3], [4], [5], [6], as a representative type of RS data, captures the reflectance spectra of surface materials across numerous narrow spectral bands. This offers fine-grained spectral features for land cover classification and demonstrates significant value in various application domains, such as surface change detection [7], vegetation monitoring [8], ecological and environmental monitoring [9], resource management [10], and geology [11]. The classification using HSI has become a key task in RS image processing [12], [13], [14]. Over the years, researchers have devoted considerable efforts to developing effective HSI feature extractors. For instance, Hong et al. [15] introduced the SpectralFormer, a transformer-driven model that captures long-range spectral relations and improves HSI classification performance. Sun et al. [16] proposed a spectral–spatial feature tokenization transformer (SSFTT) that combines 3-D/2-D convolutional spectral–spatial feature extraction with Gaussian-weighted tokenization and a transformer encoder for high-level semantic modeling, demonstrating competitive accuracy and efficiency on several benchmark HSI datasets. Zhao et al. [17] introduced GSC-ViT, a model that effectively captures local spectral–spatial information in HSI and addresses the transformer's tendency to model global correlations. However, in complex scenes, especially in urban environments, spectral characteristics of different surface materials can be very similar [18], [19]. For example, buildings and roads often exhibit no significant spectral differences, which challenges the classification in these areas using HSI. To address the limitations of HSI in complex environments, light detection and ranging (LiDAR) [20], [21], [22], [23], as another type of RS data, provides rich elevation information, due to that urban areas where objects with similar spectral characteristics, such as buildings, roads, and vegetation, can be distinguished based on their elevation differences [24]. The joint use of HSI and LiDAR significantly improves the accuracy of classification results.

With the rapid advancement of deep learning (DL), the fusion-based classification of HSI and LiDAR data has been significantly promoted, and an increasing number of DL-based methods have been proposed to enhance the performance of HSI–LiDAR fusion classification. Convolutional neural networks (CNNs) [25], [26], [27] have demonstrated strong capability in jointly learning spectral, spatial, and elevation features, thereby improving the overall accuracy (OA) of multisource classification. Hang et al. [28] introduced coupled CNNs, where two parallel branches with high-level parameters jointly learn spectral–spatial features from HSI and elevation

Received 27 January 2026; revised 21 May 2026; accepted 11 June 2026. Date of publication 15 June 2026; date of current version 29 June 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62171252 and in part by the Fundamental Research Funds for the Central Universities under Grant 00007764. *(Jiangyun Li and Jiaheng Han contributed equally to this work.) (Corresponding author: Peixian Zhuang.)*

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Digital Object Identifier 10.1109/TGRS.2026.3703805

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features from LiDAR. Lu et al. [29] proposed the coupled adversarial learning classification (CALC) framework, which extracts both modality-invariant and complementary semantic features to improve fusion accuracy. Wu et al. [30] developed the cross-channel reconstruction network (CCR-Net), a CNN-based multimodal architecture that learns compact and highly discriminative fusion representations through a cross-modal reconstruction strategy. Despite the promising performance of CNNs, their receptive field is inherently limited by the size of convolutional kernels, making it difficult to capture long-range dependencies between pixels. This hinders their ability to fully leverage global contextual information in complex HSI and LiDAR joint classification scenarios.

In response, Transformers [31], [32] have demonstrated outstanding capability in capturing long-range dependencies through the self-attention mechanism, which enables all parts of the input to be processed simultaneously and effectively models global contextual information. Yao et al. [33] proposed ExViT, an extended vision transformer for multimodal land use and land cover classification, integrating separable convolution modules and a cross-modality attention (CMA) mechanism to fuse HSI and LiDAR data effectively. Sun et al. [34] proposed M2FNet, a multiscale 3-D–2-D hybrid CNN–transformer network that employs a lightweight attention-free fusion strategy to integrate HSI and LiDAR, improving joint classification accuracy while reducing the number of parameters. Duan et al. [35] proposed the LGFR network, a dual-branch CNN–transformer model with multilevel fusion for hyperspectral and LiDAR data classification. Feng et al. [36] proposed S2EFT, a spectral–spatial–elevation fusion transformer incorporating spatial information recognition and sliding-group spectral embedding modules to address limited local modeling in traditional transformers, achieving superior classification performance on several datasets. Although transformers possess powerful representational capability, they suffer from quadratic computational complexity with respect to sequence length, leading to high computational costs for long sequences.

To overcome the limitations of transformers, Gu et al. [37] proposed the structured state space model (SSM), which achieves efficient long-sequence modeling through linear-time recurrence. Building on this foundation, Gu and Dao [38] introduced Mamba, which incorporates a selective mechanism to dynamically regulate state updates, thereby enhancing the model's contextual compression capability and improving the efficiency of long-range dependency modeling. This advancement provides a new direction for handling long-sequence processing in RS tasks. Recent studies have begun extending Mamba to multisource RS tasks. Liao et al. [39] proposed the joint classification network of HSI and LiDAR Data based on Mamba (HLMamba), which introduces a multimodal Mamba fusion module and gradient joint algorithm to enhance feature interaction and improve classification performance. Gao et al. [40] developed MSFMamba, which integrates spatial, spectral, and fusion Mamba modules to enhance HSI–LiDAR/SAR feature interaction via efficient multiscale state-space modeling. Zhang et al. [41] proposed S2CrossMamba, a state-space fusion model using cross-SSM and an inverted bottleneck cross-Mamba (IBCM) structure to refine multimodal representations and integrate spatial–spectral information. Xie et al. [42] developed DAHGMN, a hybrid GCN–Mamba network with dual-feature attention to jointly model local geometric structures and long-range dependencies, improving joint HSI and LiDAR classification. Li et al. [43], [44] proposed CMFNet and MHCMamba, two Mamba-based networks that improve HSI and LiDAR classification by enhancing multimodal feature fusion and multiscale feature extraction.

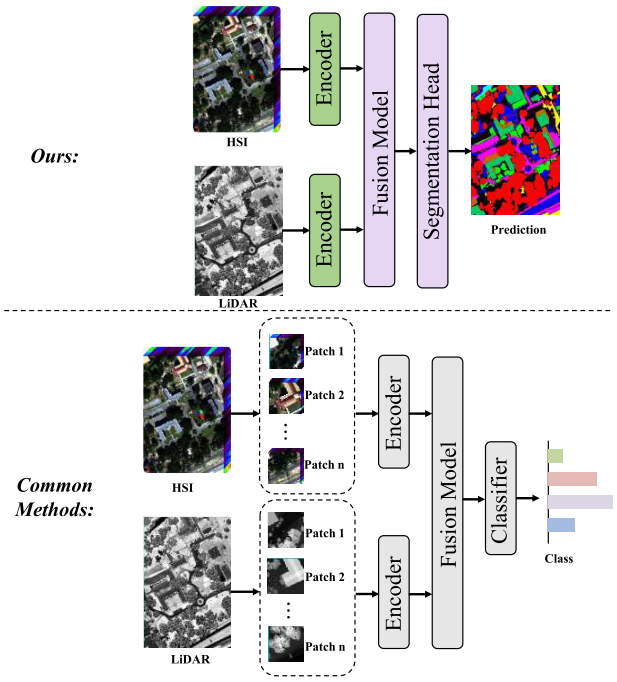


Fig. 1. (Up) Proposed WIMNet directly inputs an entire image into the model, which can reduce FLOPs and training time for more efficient semantic-level classification. (Down) Common methods divide an image into small patches for individual processing, increasing FLOPs and training time.

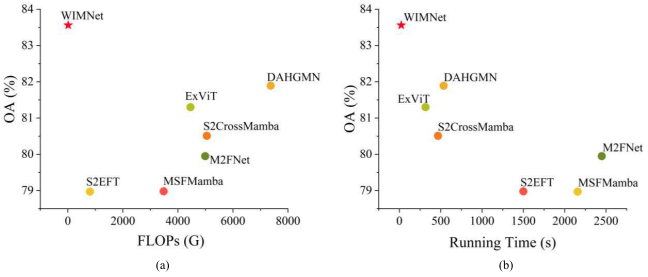


Fig. 2. Comparison between the proposed WIMNet and existing methods in terms of (a) FLOPs and (b) running time and OA on the MUUFL Dataset.

Although significant progress has been made in existing HSI and LiDAR joint classification networks, most approaches adopt a pixel-based patch processing strategy, as illustrated in Fig. 1, where an input image is divided into small patches that are individually input into the network for classification. These network architectures generally face the following limitations.

1) The patch-based processing strategy, which divides an image into small patches for individual processing, increases computational overhead, particularly for high-resolution HSI and LiDAR datasets as illustrated in

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Fig. 2, and raises the number of floating-point operations (FLOPs) and running time.

2) In existing networks, HSI features are typically extracted by independently processing spectral and spatial features before the fusion, which limits the ability to model their interdependencies and reduces the effectiveness of joint learning of spatial and spectral information.

To address the above limitations, we propose a novel whole-image-based Mamba network (WIMNet), which designs two distinct branches to extract features from HSI and LiDAR, respectively. Unlike existing patch-based HSI–LiDAR classification methods, WIMNet takes an entire image as input, thereby avoiding redundant computations caused by overlapping patches and enabling more efficient image-level semantic classification. Different from existing methods that model spectral and spatial features independently, WIMNet introduces a joint spectral–spatial Mamba branch (JSMB) to capture the interaction between spectral and spatial information. In addition, a multiscale LiDAR Mamba branch (MLMB) is designed to exploit long-range spatial dependencies and multiscale elevation features from LiDAR data. Finally, the features extracted from HSI and LiDAR are adaptively integrated by a multimodal feature fusion module (MFFM), and the classification result is obtained through a segmentation head. The main contributions of this work are summarized as follows.

1) To the best of our knowledge, the proposed WIMNet is the first image-level HSI and LiDAR joint classification network based on Mamba. Instead of segmenting an image into patch-based processing, WIMNet directly inputs an entire image for processing, which can significantly reduce computational burden, accelerate the training process, and achieve more efficient semantic-level classification.
2) We proposed the spectral-spatial mamba block (SSMB), which consists of two branches designed to extract and fuse spectral and spatial features for effective joint learning. The spectral Mamba branch extracts spectral features and employs the spectral modulation mechanism (SMM) to generate channel weights that modulate the spatial branch input, while the spatial Mamba branch performs spatial modeling.
3) Extensive experiments on three public datasets demonstrate that the proposed WIMNet outperforms 11 state-of-the-art methods in classification accuracy while achieving the lowest computational cost and running time. This confirms the superiority of our image-level Mamba architecture for joint HSI and LiDAR classification.

The remainder of the article is organized as follows. Section II provides a detailed description of the overall architecture of WIMNet and the principles behind each module. Section III presents extensive experimental results and discussions, including ablation studies, quantitative comparisons, and visual comparisons. Section IV concludes the article and highlights future research directions.

II. Preliminaries

Mamba [38] is based on the structured SSM, which combines continuous state-space equations with discrete sequence processing. It shows strong performance in modeling long sequential data. The state space represents a set of variables that can fully describe the behavior of a system. The SSM defines how these states evolve over time and predicts the next state from given inputs. The continuous form of the SSM can be expressed as follows:

$$\begin{aligned} h'(t) &= Ah(t) + Bx(t) \\ y(t) &= Ch(t) \end{aligned} \tag{1}$$

where $h(t) \in R^N$ denotes the latent state, $x(t) \in R$ represents the input signal, and $y(t) \in R$ denotes the output signal. $A \in R^{N\times N}$ is the state transition matrix, while $B \in R^N$ and $C \in R^N$ are projection matrices that map the input and latent state to the output, respectively. The latent state $h(t)$ represents the internal state of the system, which evolves over time according to the input $x(t)$ and the system dynamics determined by A.

For digital computation, the continuous SSM equations must be discretized for practical implementation. Mamba employs the zero-order hold (ZOH) technique to discretize the continuous model, enabling the transition from a continuous-time system to a discrete-time formulation. The discretization process of the state transition matrix $\tilde{A}$ and the input projection matrix $\tilde{B}$ can be expressed as follows:

$$\begin{aligned} \tilde{A} &= \exp(\Delta A) \\ \tilde{B} &= (\Delta A)^{-1}\left(\exp(\Delta A) - I\right) \cdot \Delta B \end{aligned} \tag{2}$$

where Δ denotes the discretization step size, $\tilde{A}$ represents the discrete state transition matrix, and $\tilde{B}$ denotes the discrete input matrix, while I is the identity matrix. This discretization method accurately captures the system dynamics at each time step, enabling state updates based on discrete-time intervals. The discrete form of the SSM can be expressed as follows:

$$\begin{aligned} h_t &= \tilde{A}h_{t-1} + \tilde{B}x_t \\ y_t &= Ch_t \end{aligned} \tag{3}$$

where h_t denotes the latent state at time t, y_t represents the output at time t, and C is the output matrix.

III. Proposed Methodology

A. Overall Architecture

The structure of our proposed WIMNet is illustrated in Fig. 3. Initially, the HSI and LiDAR data are processed separately through the multiscale local convolution block (MLCB) to extract local spatial features with different receptive fields. Then, WIMNet designs two distinct branches: JSMB and MLMB, to extract features from HSI and LiDAR data, respectively. The JSMB uses SSMB to extract and fuse spatial and spectral features from HSI data, while the MLMB employs Mamba, multilayer convolution, and adaptive pooling to fully extract multiscale features from LiDAR data. These features are then fused using MFFM, and the fused features are processed through a segmentation head to obtain final classification results. Table I details the definition of notations used in the proposed WIMNet.

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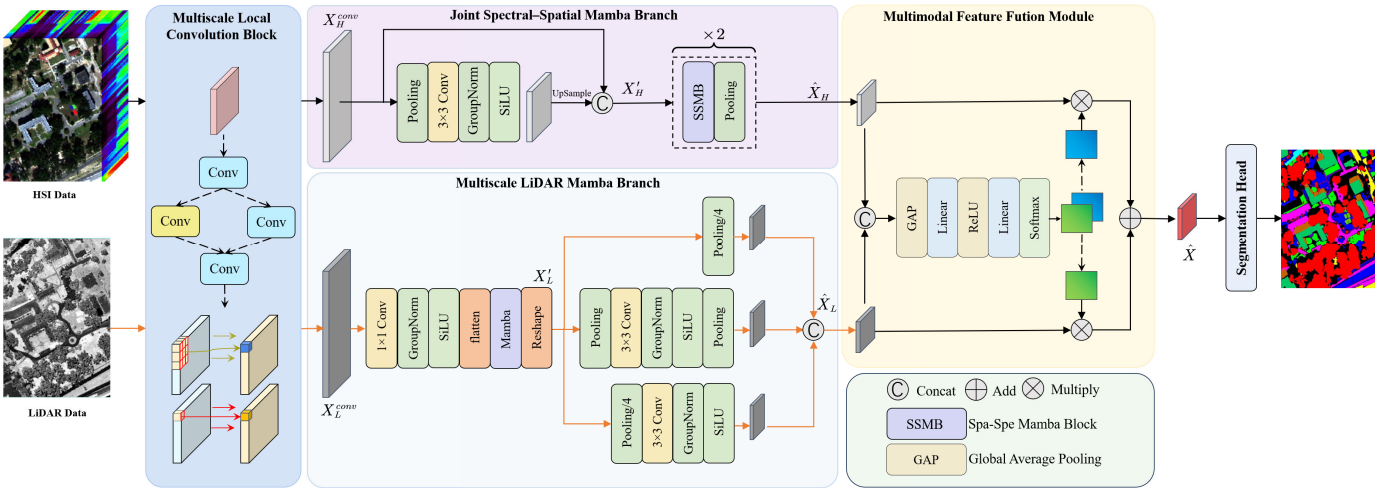


Fig. 3. Overall architecture of the proposed WIMNet. Initially, HSI and LiDAR are processed through the MLCB to extract local features. Then, JSMB and MLMB extract features from HSI and LiDAR data, respectively. Afterward, MFFM fuses the features from the two modalities, and the fused features are processed through a segmentation head to obtain final classification results.

TABLE I
DEFINITION OF NOTATIONS USED IN WIMNET

Parameters	Definition
$X_H \in \mathbb{R}^{C\times H\times W}$	The original HSI data
$X_L \in \mathbb{R}^{1\times H\times W}$	The original LiDAR data
$X_H^{\text{conv}} \in \mathbb{R}^{C_{\text{emb}}\times H\times W}$	The input of JSMB
$X_L^{\text{conv}} \in \mathbb{R}^{C_{\text{emb}}\times H\times W}$	The input of MLMB
$\hat{X}_H \in \mathbb{R}^{\frac{C_{\text{emb}}}{4}\times H\times W}$	The final HSI features output of the JSMB
$\hat{X}_L \in \mathbb{R}^{\frac{C_{\text{emb}}}{4}\times H\times W}$	The final LiDAR features output of the MLMB
$\hat{X} \in \mathbb{R}^{C_{\text{emb}}\times H\times W}$	The final fused feature
X^i	The input of the i-th SSMB ($i = 1, 2$)
Y_{spe}	The output of the spectral Mamba branch
Y_{spa}	The output of the spatial Mamba branch
Y_{fus}	The output feature of the SSMB
$\sigma(\cdot)$	The ReLU activation function
SiLU $(\cdot)$	The SiLU activation function
Sigmoid $(\cdot)$	The Sigmoid activation function
GN $(\cdot)$	The Group Normalization operation
Conv $(\cdot)$	The 1×1 convolutional layer
$\text{Conv}_1(\cdot)$	Conv + GN + SiLU
Split $(\cdot)$	The feature grouping operation
Permute $(\cdot)$	The dimension permutation operation
Flatten $(\cdot)$	The sequence flattening operation
GAP $(\cdot)$	The global average pooling operation
ω	The weight of SMM
ω_{spe}	The weight of the spectral branch
ω_{spa}	The weight of the spatial branch
ω_H	The weight of the JSMB
ω_L	The weight of the MLMB

B. Multiscale Local Convolution Block

As shown in Fig. 4, the MLCB is designed to extract local spatial features with different receptive field sizes from multimodal input data, which are embedded into a space of a specific dimension. Initially, a convolution with a kernel size of 1×1 is applied to embed the HSI data as $X_H \in \mathbb{R}^{C\times H\times W}$ or LiDAR data as $X_L \in \mathbb{R}^{1\times H\times W}$ into a specific dimensional space, where C, H and W represent the number of channels, height, and width. This process can be expressed by

$$\begin{aligned} X_H^{\text{emb}} &= \text{Conv}_1(X_H) \\ X_L^{\text{emb}} &= \text{Conv}_1(X_L) \end{aligned} \tag{4}$$

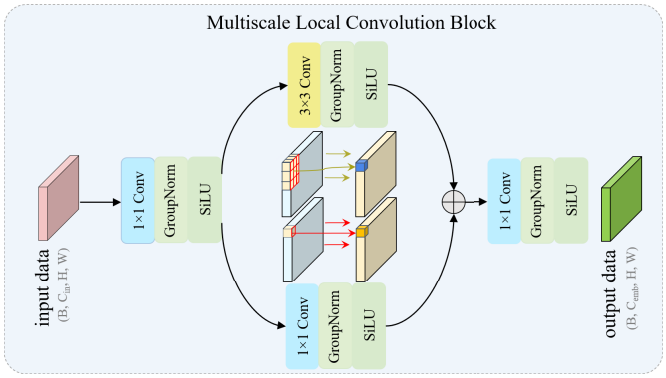


Fig. 4. Illustration of MLCB.

where X_H^{emb}, $X_L^{\text{emb}} \in \mathbb{R}^{C_{\text{emb}}\times H\times W}$ represents the embedded features, C_{emb} represents the number of embedded feature channels, which is set to 64 in our experiments. $\text{Conv}_1(\cdot)$ represents the combination of a convolution operation with a kernel size of 1×1, Group Norm layer, and SiLU activation function.

Next, parallel convolutions with kernel sizes of 3×3 and 1×1 are used to extract local spatial features with different receptive field sizes. Finally, a convolution operation is used to refine the feature representation. After each convolutional layer, a Group Norm layer and SiLU activation function are applied to help the network better adapt to the varying input data distributions and introduce nonlinear transformations. The entire process of the MLCB can be expressed as

$$\begin{aligned} X_H^{\text{conv}} &= \text{Conv}_1\left(\text{Conv}_3\left(X_H^{\text{emb}}\right) + \text{Conv}_1\left(X_H^{\text{emb}}\right)\right) \\ X_L^{\text{conv}} &= \text{Conv}_1\left(\text{Conv}_3\left(X_L^{\text{emb}}\right) + \text{Conv}_1\left(X_L^{\text{emb}}\right)\right) \end{aligned} \tag{5}$$

where X_H^{conv}, $X_L^{\text{conv}} \in \mathbb{R}^{C_{\text{emb}}\times H\times W}$ represent the final output of the MLCB.

C. Joint Spectral–Spatial Mamba Branch

The JSMB of WIMNet is designed to leverage Mamba to capture and fuse spectral and spatial features of HSI, generating a more discriminative HSI feature representation.

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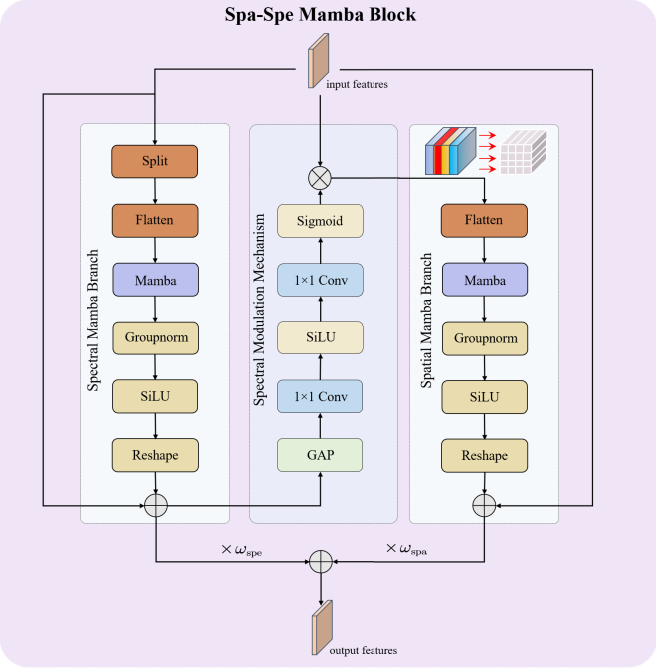


Fig. 5. Illustration of the SSMB.

1) SSMB: Existing HSI feature extraction methods extract spectral and spatial features independently and then fuse them at a later stage. Such a two-step process limits the modeling of their interdependencies and thereby constrains the effectiveness of joint learning. Considering the rich spectral information inherent in HSIs, spectral features play a crucial role in achieving accurate classification. To overcome this limitation, we propose a SSMB that facilitates joint learning through a spectral-guided spatial modeling strategy. The proposed SSMB strengthens the integration between spectral and spatial representations. As shown in Fig. 5, the SSMB consists of two parallel branches: the spectral Mamba branch and the spatial Mamba branch.

In the spectral Mamba branch, the input feature tensor is first permuted to interchange spatial and spectral dimensions. The spectral features are subsequently divided into G groups. The Mamba module is then employed to model the dependencies among these spectral groups and to update spectral representations accordingly. This process can be formulated as

$$\begin{aligned} X_{\text{split}} &= \text{Split}\left(\text{Permute}\left(X^i\right)\right) \\ X_{\text{spe}} &= \text{SiLU}\left(\text{GN}\left(\text{Mamba}\left(\text{Flatten}\left(X_{\text{split}}\right)\right)\right)\right) \\ Y_{\text{spe}} &= \text{Reshape}\left(X_{\text{spe}}\right) + X^i \end{aligned} \tag{6}$$

where X^i represents the input of SSMB ($i = 1, 2$), X_{split} represents the features obtained after channel grouping and Y_{spe} represents the final output of the spectral Mamba branch. The Mamba refers to the standard Mamba block as proposed in [38]. The Mamba block is configured with $d_{\text{model}} = C_{\text{emb}}$, $d_{\text{state}} = 16$, $d_{\text{conv}} = 4$, and expansion factor 2. The residual connection is designed to mitigate potential overfitting during training.

The resulting spectral-enhanced feature Y_{spe} is further exploited to guide spatial dependency modeling. We introduce a learnable interaction coefficient ω to obtain the guided features G_{spe} from the spectral Mamba branch. These features are then used to generate channel attention scores, which are subsequently applied to adjust the input X^i. The coefficient ω is initialized to a small value, such as 0.1, to provide a stable starting point for the interaction between spectral and spatial features, preventing overly strong coupling at the beginning of training and allowing the model to adaptively learn the appropriate interaction strength. This process can be expressed as

$$\begin{aligned} G_{\text{spe}} &= \omega Y_{\text{spe}} \\ S &= \text{Sigmoid}\left(\text{Conv}\left(\text{SiLU}\left(\text{Conv}\left(\text{GAP}\left(G_{\text{spe}}\right)\right)\right)\right)\right) \\ X^{i'} &= X^i * S \end{aligned} \tag{7}$$

where S denotes the channel attention scores obtained from the guided features, and $X^{i'}$ represents the adjusted input of the spatial Mamba branch.

The process of the spatial Mamba branch can be formulated as follows:

$$\begin{aligned} X_{\text{spa}} &= \text{SiLU}\left(\text{GN}\left(\text{Mamba}\left(\text{Flatten}\left(\text{Permute}\left(X^{i'}\right)\right)\right)\right)\right) \\ Y_{\text{spa}} &= \text{Reshape}\left(X_{\text{spa}}\right) + X^i \end{aligned} \tag{8}$$

where X_{spa} denotes the features processed by the Mamba block and Y_{spa} denotes the output of the spatial Mamba branch.

After successfully extracting spectral and spatial features, we adopt an adaptive fusion strategy to enhance their joint learning capability. The fusion process can be formulated as follows:

$$Y_{\text{fus}} = \omega_{\text{spe}} Y_{\text{spe}} + \omega_{\text{spa}} Y_{\text{spa}} + X^i \tag{9}$$

where Y_{fus} denotes the final output feature of the SSMB, and ω_{spe} and ω_{spa} represent the fusion weights for the spectral and spatial branches, respectively. These weights are optimized by backpropagation to integrate the spectral and spatial information effectively.

2) Overall Structure of JSMB: The detailed architecture of JSMB is illustrated in Fig. 3. JSMB first performs downsampling and enhancement on input features to extract more representative information. In addition, the original-scale features are incorporated as a residual connection to enrich feature diversity and strengthen the representation capability. This design provides more robust features for subsequent SSMBs. Two consecutive SSMBs are then employed to extract discriminative HSI features. The final extracted HSI features $\hat{X}_H$ can be obtained as follows:

$$\begin{aligned} X'_H &= \text{Up}\left(\text{Conv}_3\left(\text{Pooling}_{1/2}\left(X_H^{\text{conv}}\right)\right)\right) + X_H^{\text{conv}} \\ \hat{X}_H &= \text{Pooling}_{1/2}\left(\text{SSMB}\left(\text{Pooling}_{1/2}\left(\text{SSMB}\left(X'_H\right)\right)\right)\right) \end{aligned} \tag{10}$$

where Up$(\cdot)$ denotes upsampling using the bilinear interpolation, and SSMB$(\cdot)$ represents the SSMB. $X'_H \in \mathbb{R}^{C_{emb} \times H \times W}$ and $\hat{X}_H \in \mathbb{R}^{(C_{emb})/4 \times H \times W}$ denote input features of the SSMB and final HSI features output of the JSMB, respectively. The pooling operation reduces feature map size, thereby lowering subsequent computational cost while preserving essential features.

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D. Multiscale LiDAR Mamba Branch

The MLMB is designed to model the spatial dependencies within LiDAR data using Mamba, while simultaneously integrating multiscale spatial information to extract more representative spatial features. The detailed architecture is shown in Fig. 3. In this branch, a convolutional operation is first applied to reduce the channel dimension of the LiDAR input by half, thereby lowering computational complexity and yielding more compact spatial representations. Global contextual information is then captured using the Mamba module. This process is formulated as

$$X_L' = \text{Reshape}\left(\text{Mamba}\left(\text{Flatten}\left(\text{Conv}_1\left(X_L^{\text{conv}}\right)\right)\right)\right) \tag{11}$$

where $X_L' \in \mathbb{R}^{C_{\text{emb}} \times H \times W}$ denotes the features obtained after Mamba processing. Subsequently, MLMB performs multiscale enhancement to integrate spatial information across different scales. It extracts and aligns spatial features from multiple resolutions to generate LiDAR features with stronger representational capacity. To this end, the adaptive pooling is applied to reduce the spatial resolution by a factor of 4 obtaining a base feature representation X_L^1 while downsampling and convolutional operations are used to extract spatial context features X_L^2 and X_L^3 at different scales. The resulting multiscale features are concatenated and passed through a convolutional layer for channel alignment, producing final spatially enriched LiDAR features, which are expressed as follows:

$$\begin{aligned} X_L^1 &= \text{Pooling}_{1/4}\left(X_L'\right) \\ X_L^2 &= \text{Conv}_3\left(\text{Pooling}_{1/4}\left(X_L'\right)\right), \\ X_L^3 &= \text{Pooling}_{1/2}\left(\text{Conv}_3\left(\text{Pooling}_{1/2}\left(X_L'\right)\right)\right) \\ \hat{X}_L &= \text{Conv}\left(\text{Concat}\left[X_L^1, X_L^2, X_L^3\right]\right) \end{aligned} \tag{12}$$

where $X_L^1 \in \mathbb{R}^{(C_{emb})/4 \times H \times W}$, $X_L^2 \in \mathbb{R}^{(C_{emb})/4 \times H \times W}$, and $X_L^3 \in \mathbb{R}^{(C_{emb})/4 \times H \times W}$ denote the base and multiscale spatial features, respectively, and $\hat{X}_L \in \mathbb{R}^{(C_{emb})/4 \times H \times W}$ represents the final output of the MLMB.

E. Multimodal Feature Fusion Module

Effective feature fusion is critical for fully leveraging the complementarity between HSI and LiDAR data, thereby enhancing classification performance. To this end, we design a MFFM, which adaptively generates weights based on the joint representation of HSI and LiDAR features, enabling an efficient integration of the two modalities. Specifically, the HSI and LiDAR features extracted by JSMB and MLMB are first concatenated along the channel dimension as the first step of fusion. A global average pooling operation is then applied to obtain the global context feature $\hat{X}_G$. This is followed by two fully connected layers to generate attention weights ω_H and ω_L corresponding to each modality. The final fused feature $\hat{X}$ is obtained through the following process:

$$\begin{aligned} \hat{X}_G &= \text{GAP}\left(\text{Concat}\left[\hat{X}_H, \hat{X}_L\right]\right) \\ [\omega_H, \omega_L] &= \text{Softmax}\left(W_2 \sigma\left(W_1 X_G\right)\right) \\ \hat{X} &= \omega_H \hat{X}_H + \omega_L \hat{X}_L \end{aligned} \tag{13}$$

where $\hat{X}_G \in \mathbb{R}^{2C_{\text{emb}} \times H \times W}$ denotes the feature obtained through the global average pooling, W_1 and W_2 represent the learnable weight matrices of the two fully connected layers, ω_H and ω_L denote attention weights for the JSMB and MLMB, respectively, and $\hat{X} \in \mathbb{R}^{C_{\text{emb}} \times H \times W}$ represents the final fused feature.

Algorithm 1 Pseudo Procedure of the Proposed WIMNet

Input : HSI data $X_H \in \mathbb{R}^{C \times H \times W}$, LiDAR data $X_L \in \mathbb{R}^{1 \times H \times W}$

Output: Predicted labels $Pred$.

1. Multiscale Local Convolution Block:

$X_H^{\text{emb}} = \text{Conv}_1(X_H)$

$X_L^{\text{emb}} = \text{Conv}_1(X_L)$

$X_H^{\text{conv}} = \text{Conv}_1\left(\text{Conv}_3(X_H^{\text{emb}}) + \text{Conv}_1(X_H^{\text{emb}})\right)$

$X_L^{\text{conv}} = \text{Conv}_1\left(\text{Conv}_3(X_L^{\text{emb}}) + \text{Conv}_1(X_L^{\text{emb}})\right)$

2. Joint Spectral–Spatial Mamba Branch:

1) Downsampling with Residual Connection:

$X_H' = \text{Up}\left(\text{Conv}_{3\times3}\left(\text{Pooling}_{1/2}(X_H^{\text{conv}})\right)\right) + X_H^{\text{conv}}$

2) Two Spa-Spe Mamba Blocks with Intermediate Pooling:

$X = X_H'$

for *each stage* $i = 1, 2$ **do**

$X_{\text{split}} = \text{Split}\left(\text{Permute}(X)\right)$

$X_{\text{spe}} = \text{SiLU}\left(\text{GN}\left(\text{Mamba}\left(\text{Flatten}(X_{\text{split}})\right)\right)\right)$

$Y_{\text{spe}} = \text{Reshape}(X_{\text{spe}}) + X$

$G_{\text{spe}} = \omega \cdot Y_{\text{spe}}$

$S = \text{Sigmoid}\left(\text{Conv}_{1\times1}\left(\text{SiLU}\left(\text{Conv}_{1\times1}\left(\text{GAP}(G_{\text{spe}})\right)\right)\right)\right)$

$X' = X \cdot S$

$X_{\text{spa}} = \text{SiLU}\left(\text{GN}\left(\text{Mamba}\left(\text{Flatten}\left(\text{Permute}(X')\right)\right)\right)\right)$

$Y_{\text{spa}} = \text{Reshape}(X_{\text{spa}}) + X$

$X = \omega_{\text{spe}} Y_{\text{spe}} + \omega_{\text{spa}} Y_{\text{spa}} + X$

$X = \text{Pooling}_{1/2}(X)$ (performed after each SSMB)

$\hat{X}_H = X$

3. Multiscale LiDAR Mamba Branch:

$X_L' = \text{Reshape}\left(\text{Mamba}\left(\text{Flatten}\left(\text{Conv}_1(X_L^{\text{conv}})\right)\right)\right)$

$X_L^1 = \text{Pooling}_{1/4}\left(X_L'\right)$

$X_L^2 = \text{Conv}_3\left(\text{Pooling}_{1/4}(X_L')\right)$

$X_L^3 = \text{Pooling}_{1/2}\left(\text{Conv}_3\left(\text{Pooling}_{1/2}(X_L')\right)\right)$

$\hat{X}_L = \text{Conv}\left(\text{Concat}\left[X_L^1, X_L^2, X_L^3\right]\right)$

4. Multimodal Feature Fusion Module:

$\hat{X}_G = \text{GAP}\left(\text{Concat}\left[\hat{X}_H, \hat{X}_L\right]\right)$

$[\omega_H, \omega_L] = \text{Softmax}\left(W_2 \cdot \sigma\left(W_1 \cdot \hat{X}_G\right)\right)$

$\hat{X} = \omega_H \cdot \hat{X}_H + \omega_L \cdot \hat{X}_L$

5. Classifier:

$\hat{X}_{up} = \text{Up}(\hat{X})$

$X_{seg} = \text{SiLU}\left(\text{GN}\left(\text{Conv}^{C_{\text{emb}} \to 128}(\hat{X}_{up})\right)\right)$

$Pred = \text{Conv}^{128 \to N_c}(X_{seg})$

F. Segmentation Head

After multimodal feature fusion, the fused feature $\hat{X}$ is fed into a lightweight segmentation head to generate the final classification map. Specifically, $\hat{X}$ is first upsampled to

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the original spatial resolution by bilinear interpolation. Then, two 1×1 convolutional layers are used to perform channel projection and pixel-wise classification. The process can be formulated as

$$\begin{aligned} \hat{X}_{up} &= \mathrm{Up}\left(\hat{X}\right) \\ X_{seg} &= \mathrm{SiLU}\left(\mathrm{GN}\left(\mathrm{Conv}^{C_{\mathrm{emb}}\to 128}\left(\hat{X}_{up}\right)\right)\right) \\ Y &= \mathrm{Conv}^{128\to N_c}\left(X_{seg}\right) \end{aligned} \tag{14}$$

where $\hat{X}_{up}$ and X_{seg} denote the upsampled feature and intermediate projected feature, respectively. N_c is the number of classes and $Y \in \mathbb{R}^{N_c \times H \times W}$ denotes the final segmentation logits. The overview of the inference process of WIMNet is summarized in Algorithm 1 in the form of pseudocode.

G. Loss Function

In order to improve the stability of optimization and enhance model generalization, the proposed loss function refines the traditional fusion loss by incorporating label smoothing [45], [46] into the standard cross-entropy formulation [47]. By employing a smoothed target distribution to regularize the training process, the method effectively mitigates overconfidence and improves convergence robustness. Label smoothing replaces the one-hot target distribution with a softened version, which prevents the model from being overconfident in its predictions and enhances the stability of optimization. The LSCE loss can be reformulated in a simplified form as

$$\mathcal{L}_{LSCE} = -\frac{1}{N}\sum_{i=1}^{N}\left[(1-\varepsilon)\log p_{i,y_i} + \frac{\varepsilon}{C}\sum_{c=1}^{C}\log p_{i,c}\right]. \tag{15}$$

where p_{i,y_i} denotes the predicted probability of the ground-truth class for the ith sample, C is the total number of classes, and ε is the smoothing factor that controls the regularization strength. When $\varepsilon = 0$, (15) reduces to the standard cross-entropy loss. This design encourages the network to learn smoother and more generalizable class distributions, alleviating overfitting and improving convergence stability.

IV. Experiment

To validate the effectiveness of the proposed WIMNet, numerous experiments are conducted on three HSI-LiDAR datasets: Houston2013, MUUFL, and Trento. Initially, the descriptions of the three datasets used in the experiments are detailed. Subsequently, the experimental setup and evaluation metrics are presented. Following this, a series of experimental comparisons are carried out between WIMNet and 11 state-of-the-art methods, including quantitative comparisons, visual comparisons, and comparisons of model computational cost and time performance. Finally, the ablation experiments of the proposed method are presented.

A. Data Description

1) Houston2013 Dataset [48]*:* Collected in 2013 over the Houston urban area using the compact airborne spectral imager (CASI) sensor, this multimodal dataset integrates HSI and LiDAR-derived digital surface model (DSM) data. The HSI data includes 144 spectral bands ranging from 0.38 to 1.35 μm, with a spatial resolution of 2.5 m and a spatial extent of 345×1905 pixels. The dataset includes 15 029 labeled samples across 15 land cover categories, representing diverse urban environments. The combination of HSI and LiDAR data allows for detailed analysis of urban land cover, with challenges of spectral confusion and within-class variability.

2) MUUFL Dataset [49]*:* Acquired in November 2010 over the Gulfport campus of the University of Southern Mississippi, this dataset combines HSI data with LiDAR-derived DSM information. The HSI data, captured by a CASI-1500 spectrometer, includes 64 spectral bands from 0.38 to 1.05 μm, with a spatial resolution of 0.54×1.0 m. The LiDAR data, obtained using an ALTM system at 1064 nm, has a spatial resolution of 0.60×0.78 m. The dataset consists of 53 687 labeled samples across 11 land cover categories, representing coastal and urban environments, and provides insights into spectral and structural variations in these areas.

3) Trento Dataset: Collected over rural areas south of Trento, Italy, this dataset integrates HSI and LiDAR data. The HSI data contains 63 spectral bands (0.42–0.99 μm), with a spatial resolution of 1 m. The LiDAR data, captured by a 3100EA sensor, provides two height channels. The dataset spans 600×166 pixels and includes 30 214 labeled samples across six land cover categories, capturing spectral and spatial characteristics of Mediterranean agro-forest ecosystems.

The detailed numbers of the training and testing samples for each dataset are presented in Table II.

B. Experimental Setup

1) Implementation Details: All experiments in this study were implemented using the PyTorch framework and conducted on an NVIDIA GeForce RTX 4090 GPU with 24 GB of memory. For each dataset, 20 samples per class were randomly selected as training data, while the remaining samples were used for testing. The proposed WIMNet was optimized using the Adam optimizer, with the number of training epochs set to 200 and the group number G set to 4. The learning rate was set to 0.0005 for the Houston2013 dataset and 0.001 for the MUUFL and Trento datasets.

2) Evaluation Metrics: To assess the effectiveness of the proposed method in joint hyperspectral and LiDAR classification, three widely used quantitative indicators are employed: OA, average accuracy (AA), and the kappa coefficient (κ). OA measures the proportion of correctly predicted samples over the entire dataset, providing a global view of classification performance. AA represents the mean of the individual class accuracies, thereby reflecting the model's balance across different categories. The κ coefficient evaluates the consistency between the predicted map and the reference labels by accounting for chance agreement.

3) Comparison Methods: To validate the effectiveness of the proposed WIMNet, 11 representative methods are selected for comparison: SF [15], SSFTT [16], CALC [29], CoupledCNNs [28], CCR-Net [30], M2FNet [34], ExViT [33], S2EFT [36], DAHGMN [42], S2CrossMamba [41], and MSFMamba [40]. Among them, both SF and SSFTT belong to transformer-based HSI single-source classification methods, while the

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TABLE II

DETAILS OF EACH LAND-COVER CLASS OF THREE DATASETS USED FOR EVALUATION IN THE EXPERIMENTS

Houston2013				**MUUFL**				**Trento**			
No.	Name	Train.	Test.	No.	Name	Train.	Test.	No.	Name	Train.	Test.
C1	Healthy Grass	20	1231	C1	Trees	20	23226	C1	Apple Trees	20	4014
C2	Stressed Grass	20	1234	C2	Mostly Grass	20	4250	C2	Buildings	20	2883
C3	Synthetic Grass	20	677	C3	Mixed Ground Surface	20	6862	C3	Ground	20	459
C4	Trees	20	1224	C4	Dirt and Sand	20	1806	C4	Woods	20	9103
C5	Soil	20	1222	C5	Road	20	6667	C5	Vineyard	20	10481
C6	Water	20	305	C6	Water	20	446	C6	Roads	20	3154
C7	Residential	20	1248	C7	Buildings Shadow	20	2213	-	-	-	-
C8	Commercial	20	1224	C8	Buildings	20	6220	-	-	-	-
C9	Road	20	1232	C9	Sidewalk	20	1365	-	-	-	-
C10	Highway	20	1207	C10	Yellow Curb	20	163	-	-	-	-
C11	Railway	20	1215	C11	Cloth Panels	20	249	-	-	-	-
C12	Parking Lot 1	20	1213	-	-	-	-	-	-	-	-
C13	Parking Lot 2	20	449	-	-	-	-	-	-	-	-
C14	Tennis Court	20	408	-	-	-	-	-	-	-	-
C15	Running Track	20	640	-	-	-	-	-	-	-	-
Total	-	300	14729	-	-	220	53467	-	-	120	30094

TABLE III

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE HOUSTON2013 DATASET

Class	HSI		HSI+LiDAR									
	Transformer-based		CNN-based			Transformer-based			Mamba-based			
	SF	SSFTT	CALC	Coupled-CNNs	CCR-Net	M2FNet	ExViT	S2EFT	DAHGMN	S2CrossMamba	MSFMamba	WIMNet
	TGRS2021	TGRS2022	IF2022	TGRS2020	TGRS2022	TGRS2024	TGRS2023	KBS2024	TGRS2025	TGRS2024	TGRS2025	Ours
1	81.07	94.80	90.41	88.54	86.43	84.13	94.88	97.40	83.51	95.61	95.29	**99.68**
2	97.81	95.46	98.06	85.25	96.11	95.24	96.11	95.30	**99.35**	93.76	91.90	97.00
3	95.13	96.31	97.49	**100.00**	94.68	98.29	99.70	99.56	99.70	98.38	**100.00**	99.85
4	88.97	**96.90**	98.04	95.26	96.08	95.68	94.04	96.81	**96.90**	96.32	87.09	92.24
5	96.40	**100.00**	**100.00**	85.02	92.72	99.13	**100.00**	98.69	99.92	100.00	**100.00**	**100.00**
6	83.93	**100.00**	**100.00**	91.80	85.57	95.81	99.67	87.87	**100.00**	96.39	98.03	99.34
7	69.23	93.75	97.44	72.19	82.77	92.22	98.24	87.90	93.67	**98.32**	81.49	95.19
8	54.17	74.10	86.27	83.91	94.44	90.90	88.56	79.90	**94.61**	87.91	90.64	87.83
9	64.20	78.81	75.73	28.33	78.65	92.35	93.91	71.35	90.99	76.46	94.89	**95.62**
10	48.30	94.95	97.43	81.28	59.57	76.32	86.91	78.62	88.65	95.19	81.11	**99.42**
11	30.04	99.75	97.37	94.73	37.45	94.44	89.47	85.27	**100.00**	99.17	86.09	85.93
12	60.68	93.16	93.57	60.02	51.77	94.56	89.86	60.51	89.37	85.90	82.93	**96.46**
13	59.02	92.87	95.99	87.31	88.20	91.15	96.66	67.48	95.55	**99.11**	95.55	94.43
14	97.30	**100.00**	99.75	96.08	91.18	99.35	99.75	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
15	76.09	**100.00**	**100.00**	86.72	97.66	98.26	**100.00**	99.84	99.53	99.53	**100.00**	99.22
OA(%)	71.44	93.09	94.28	79.98	80.21	92.26	94.24	86.13	93.81	93.94	91.79	**95.57**
AA(%)	73.49	94.06	95.17	82.43	82.22	93.18	95.18	87.10	94.82	94.86	93.13	**96.15**
$\kappa\times$ 100	69.17	92.53	93.82	78.40	78.59	91.61	93.77	85.35	93.31	93.45	91.12	**95.21**

TABLE IV

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE MUUFL GULFPORT DATASET

Class	HSI		HSI+LiDAR									
	Transformer-based		CNN-based			Transformer-based			Mamba-based			
	SF	SSFTT	CALC	Coupled-CNNs	CCR-Net	M2FNet	ExViT	S2EFT	DAHGMN	S2CrossMamba	MSFMamba	WIMNet
	TGRS2021	TGRS2022	IF2022	TGRS2020	TGRS2022	TGRS2024	TGRS2023	KBS2024	TGRS2025	TGRS2024	TGRS2025	Ours
1	75.37	**95.06**	92.91	87.57	91.90	87.89	87.26	89.30	86.86	91.44	83.56	90.94
2	69.06	59.13	65.53	76.33	60.52	74.28	67.79	66.31	82.87	68.71	**85.11**	70.24
3	48.85	57.43	61.47	58.93	54.75	60.94	54.63	51.50	61.28	54.04	54.47	**71.01**
4	62.62	72.65	79.90	80.07	74.47	81.89	95.02	74.69	69.71	77.46	79.67	**94.85**
5	77.41	55.74	75.19	**84.04**	80.44	73.19	82.29	77.80	78.03	71.92	78.88	79.92
6	68.16	99.78	100.00	97.76	98.43	99.78	**100.00**	96.19	**100.00**	**100.00**	99.55	99.78
7	66.43	**89.25**	79.39	76.82	70.85	75.28	85.86	72.53	91.78	78.90	75.19	85.81
8	36.67	87.28	**97.01**	95.80	89.53	89.50	94.52	92.91	93.96	92.15	90.11	88.34
9	49.74	45.71	37.51	**73.41**	61.90	45.49	56.19	33.48	54.14	50.33	49.30	49.30
10	73.62	77.91	**84.66**	76.07	64.42	60.74	82.82	60.12	76.69	84.05	81.59	76.07
11	93.98	**99.60**	**99.60**	95.98	97.59	95.98	97.59	91.97	95.58	95.18	**99.60**	98.80
OA(%)	65.78	79.32	82.62	82.55	80.70	79.95	81.30	78.97	81.89	80.51	78.98	**83.98**
AA(%)	65.63	76.32	79.38	82.07	76.80	76.82	82.18	73.34	80.99	78.56	79.73	**82.28**
$\kappa\times$ 100	57.15	72.61	77.24	77.44	74.54	74.07	76.03	72.68	76.77	74.74	73.14	**79.20**

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TABLE V
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE TRENTO DATASET

Class	HSI		HSI+LiDAR									
	Transformer-based		CNN-based			Transformer-based			Mamba-based			
	SF	SSFTT	CALC	Coupled-CNNs	CCR-Net	M2FNet	ExViT	S2EFT	DAHGMN	S2CrossMamba	MSFMamba	WIMNet
	TGRS2021	TGRS2022	IF2022	TGRS2020	TGRS2022	TGRS2024	TGRS2023	KBS2024	TGRS2025	TGRS2024	TGRS2025	Ours
1	98.23	99.03	97.46	97.23	97.91	99.23	97.04	96.76	97.46	93.82	98.56	**99.88**
2	88.00	97.50	**99.79**	96.91	96.81	99.20	98.79	94.28	95.04	99.69	95.69	98.99
3	98.69	**100.00**	97.39	93.46	94.34	**100.00**	**100.00**	94.77	98.91	97.17	94.34	96.51
4	81.91	**100.00**	**100.00**	98.91	99.91	99.59	99.97	99.98	**100.00**	**100.00**	99.99	**100.00**
5	93.67	**100.00**	**100.00**	**100.00**	98.61	99.89	99.79	99.01	**100.00**	99.80	95.48	99.97
6	71.21	87.00	95.47	91.98	**96.70**	91.82	95.09	88.49	93.50	96.42	96.95	95.72
OA(%)	87.90	98.27	99.13	98.07	98.47	98.80	98.89	97.38	98.49	98.66	97.41	**99.38**
AA(%)	88.62	97.26	98.35	96.42	97.38	98.29	98.44	95.55	97.86	97.82	96.83	**98.51**
$\kappa\times$ 100	84.11	97.69	98.83	98.92	97.96	98.40	98.52	96.50	97.98	98.21	96.56	**99.17**

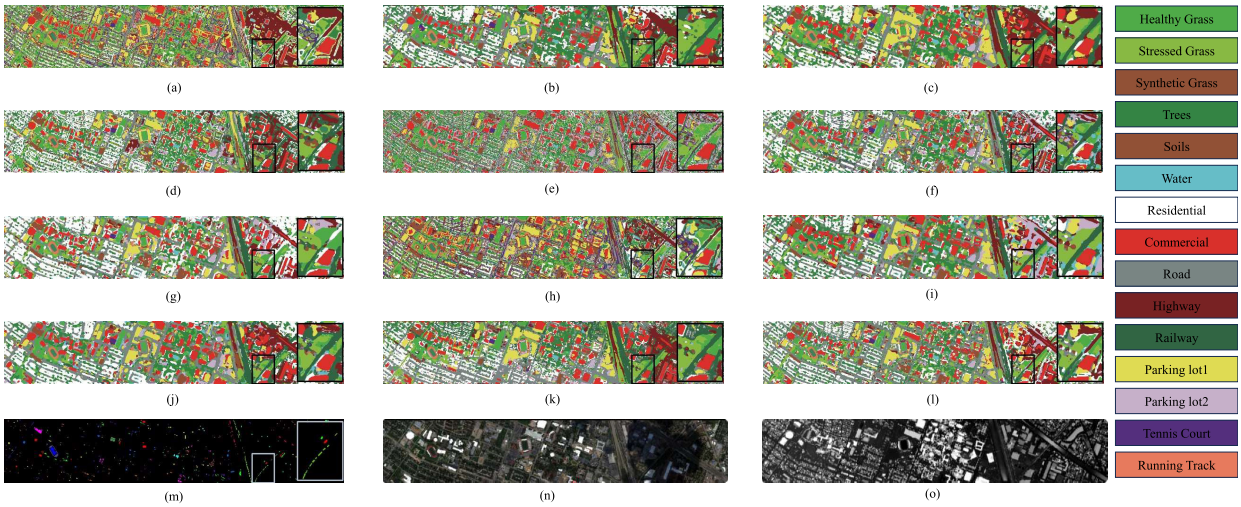


Fig. 6. Classification result maps on the Houston2013 dataset. (a) SF (71.44%). (b) SSFTT (93.09%). (c) CALC (94.28%). (d) Coupled-CNNs (79.98%). (e) CCR-Net (80.21%). (f) M2FNet (92.26%). (g) ExViT (94.24%). (h) S2EFT (86.13%). (i) DAHGMN (93.81%). (j) S2CrossMamba (93.94%). (k) MSFMamba (91.79%). (l) WIMNet (95.57%). (m) Ground-truth. (n) HSI pseudo-color map. (o) LiDAR-based DSM.

remaining nine methods perform joint classification using both HSI and LiDAR data. Specifically, CALC, CoupledCNNs, and CCR-Net are CNN-based architectures; M2FNet, ExViT, and S2EFT are transformer-based models; while DAHGMN, S2CrossMamba, and MSFMamba are Mamba-based models. The hyperparameter settings for the comparison methods are consistent with those reported in the original papers.

C. Comparative Experiments

1) Quantitative Analysis: Tables III–V report the quantitative classification results of various methods on the three datasets. Each table shows the per-class accuracy, OA, AA, and Kappa. It is noted that the proposed WIMNet consistently achieves superior or competitive results across all datasets, obtaining the highest OA and Kappa values in all cases. Moreover, the results clearly show that introducing LiDAR data notably enhances the classification performance compared to single-modality HSI methods (e.g., SF and SSFTT), demonstrating the advantage of multimodal fusion for spectral–spatial representation learning. For the Houston2013 dataset (Table III), our WIMNet achieves superior performance with an OA of 95.57%, an AA of 96.15%, and a Kappa of 95.21, all of which exceed those of other competing approaches. Specifically, compared with CALC, which achieves the highest OA (94.28%) and Kappa (93.82) among the compared methods, our WIMNet improves OA by 1.29% and Kappa by 1.39. Compared to ExViT, which attains the highest AA (95.18%), our WIMNet improves AA by 0.97%. In addition, our WIMNet achieves the highest accuracy in five out of the 15 categories on the Houston dataset, including challenging classes such as road, highway, and parking lot 1. For the MUUFL dataset (Table IV), our WIMNet achieves an OA of 83.98%, an AA of 82.28%, and a Kappa of 79.20. Compared to CALC, which achieves the highest OA (82.62%) among the compared methods, our WIMNet improves OA by 1.36%. Compared to ExViT, which attains the highest AA (82.18), our WIMNet improves AA by 0.1%. Similarly, compared to Coupled-CNNs, which attains the highest Kappa (77.44), our WIMNet surpasses it by 1.76 in Kappa. In addition, WIMNet achieves the highest accuracy in two categories, showing clear advantages in handling mixed classes, including mixed ground surface and dirt. In the Trento dataset (Table V), our WIMNet achieves an OA of 99.38%, an AA of 98.51, and a Kappa of 99.17, outperforming all compared methods. Compared with CALC, which achieves

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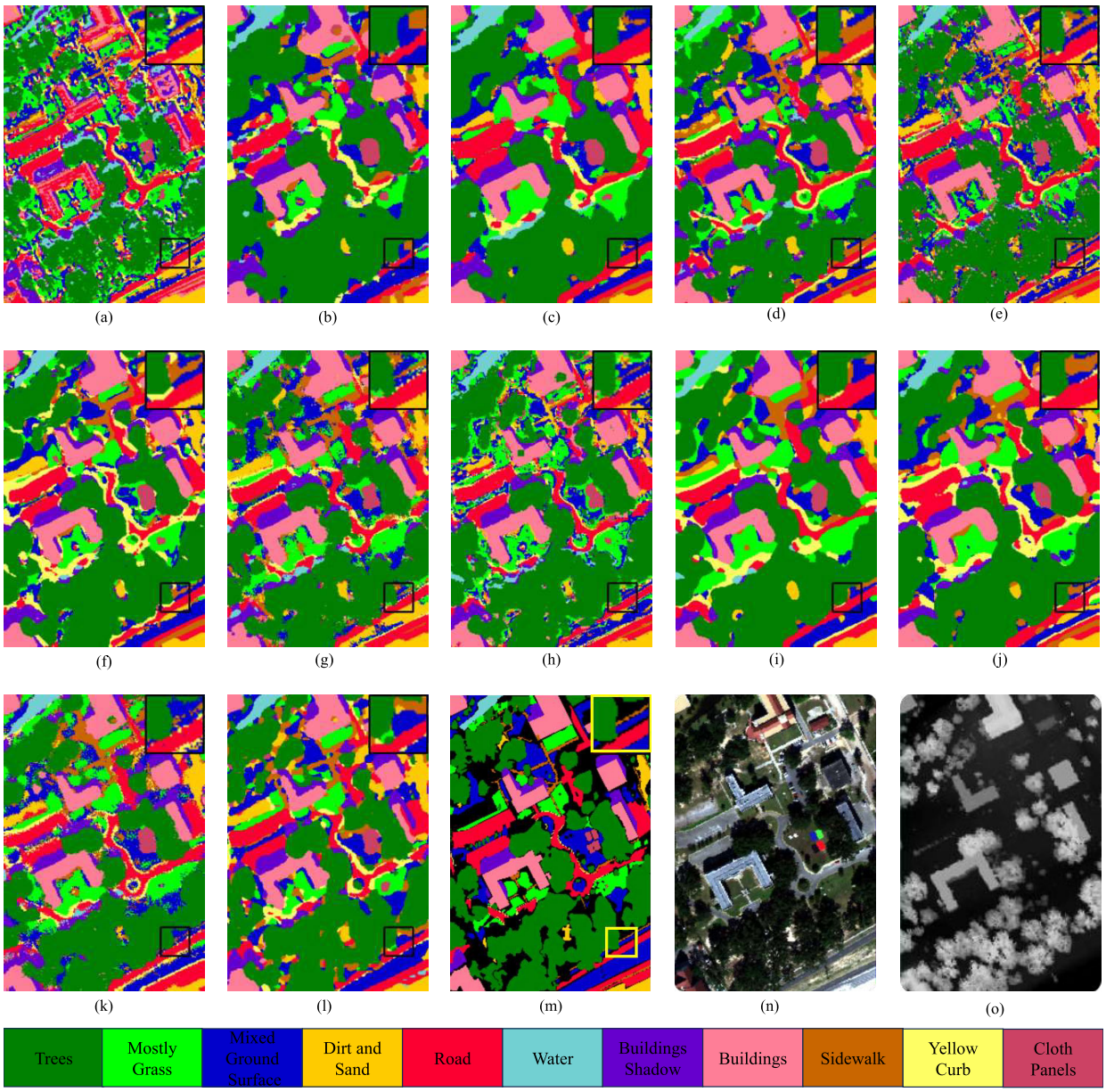


Fig. 7. Classification result maps on the MUUFL dataset. (a) SF (65.78%). (b) SSFTT (79.32%). (c) CALC (82.62%). (d) Coupled-CNNs (82.55%). (e) CCR-Net (80.70%). (f) M2FNet (79.95%). (g) ExViT (81.30%). (h) S2EFT (78.97%). (i) DAHGMN (81.89%). (j) S2CrossMamba (80.51%). (k) MSFMamba (78.98%). (l) WIMNet (83.98%). (m) Ground-truth. (n) HSI pseudo-color map. (o) LiDAR-based DSM.

the highest OA (99.13%), AA (98.35%), and Kappa (99.02) among the compared methods, WIMNet improves OA by 0.25%, AA by 0.26%, and Kappa by 0.15. Compared to Coupled-CNNs, which attain the highest Kappa (98.92), our WIMNet improves Kappa by 0.25. Across all datasets, our WIMNet achieves the highest OA and Kappa values, outperforming CNN-based, transformer-based, and Mamba-based approaches and maintaining competitive AA performance. The consistent superiority verifies the effectiveness of the proposed multimodal hierarchical fusion strategy. Notably, incorporating LiDAR information leads to notable improvements compared with the best-performing HSI single-source classification method, yielding an OA improvement of over 1.1% across the three datasets. These results demonstrate that WIMNet achieves state-of-the-art performance and offers a reliable framework for joint HSI and LiDAR classification.

2) Visual Evaluation: Figs. 6–8 show the classification maps of different methods across the three datasets. Overall, WIMNet produces cleaner and more structured maps, with better spatial consistency and fewer misclassifications. In the Houston2013 dataset, the zoomed-in region of Fig. 6 focusing on the railway class shows that our WIMNet produces results highly consistent with the ground truth. In the MUUFL dataset, the zoomed-in region of Fig. 7 focusing on the road and sidewalk classes shows that our WIMNet produces results highly consistent with the ground truth. In the Trento dataset, the zoomed-in region of Fig. 8 focusing on the buildings and roads classes shows that our WIMNet produces results highly

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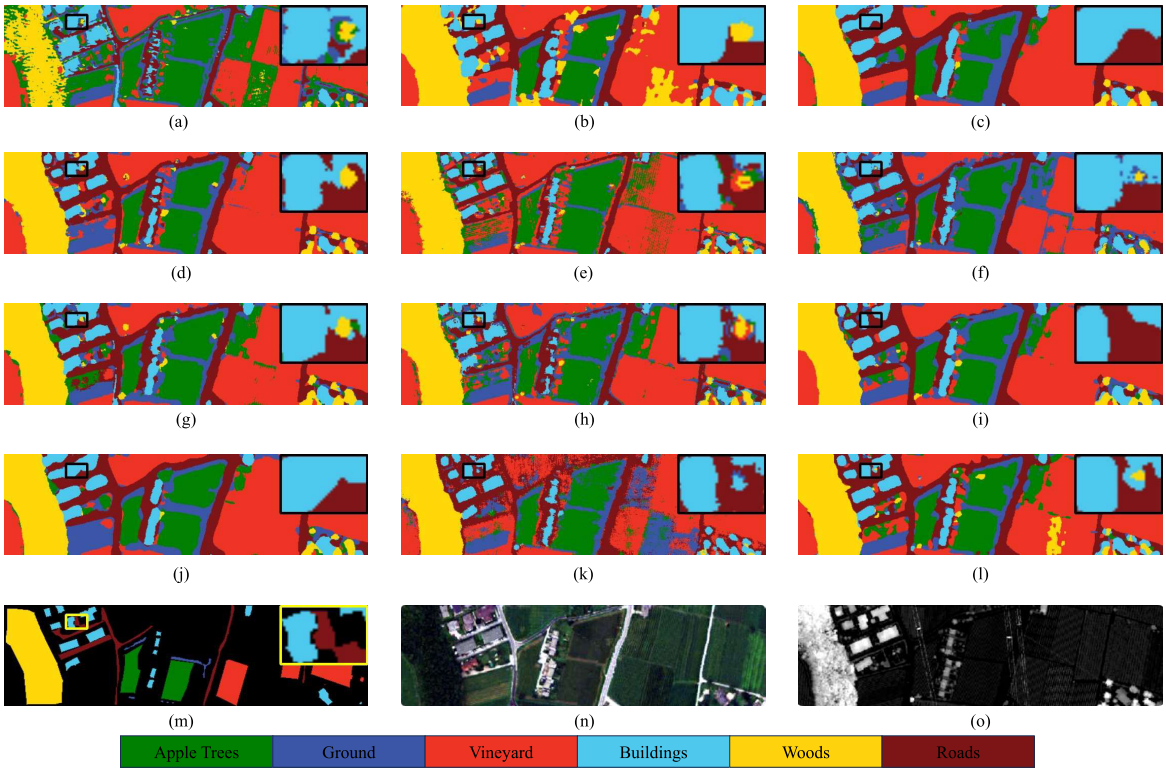


Fig. 8. Classification result maps on the Trento dataset. (a) SF (87.90%). (b) SSFTT (98.27%). (c) CALC (99.13%). (d) Coupled-CNNs (98.07%). (e) CCR-Net (98.47%). (f) M2FNet (98.80%). (g) ExViT (98.89%). (h) S2EFT (97.38%). (i) DAHGMN (98.49%). (j) S2CrossMamba (98.66%). (k) MSFMamba (97.41%). (l) WIMNet (99.38%). (m) Ground-truth. (n) HSI pseudo-color map. (o) LiDAR-based DSM.

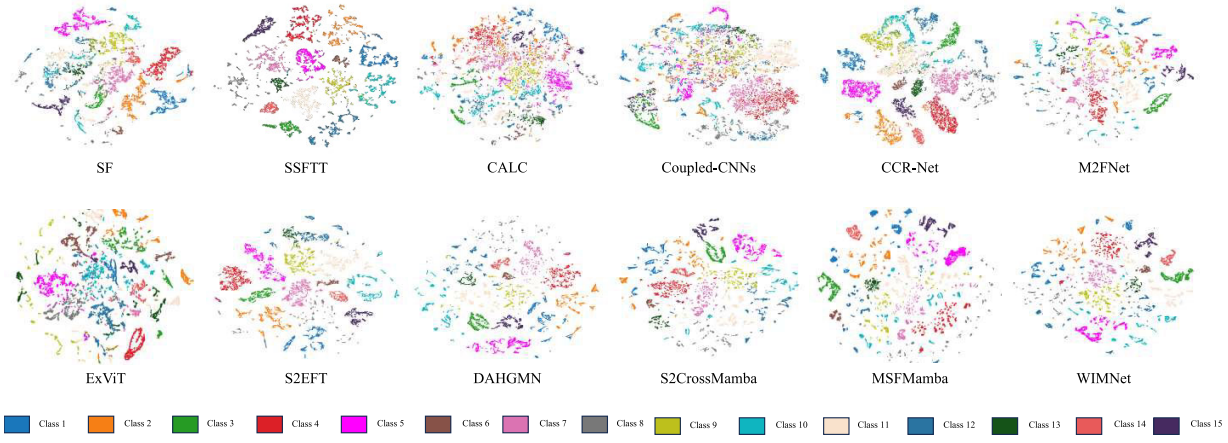


Fig. 9. t-SNE visualization on the Houston2013 dataset.

TABLE VI

COMPUTATIONAL COST AND RUNTIME OF THE PROPOSED WIMNET VERSUS COMPARISON METHODS

Dataset	Metric	M2FNet	ExViT	S2EFT	DAHGMN	S2CrossMamba	MSFMamba	WIMNet
Houston2013	Train time (s)	816.20	390.81	1090.90	353.78	234.60	1848.80	**201.27**
	Test time (s)	16.34	4.42	5.15	15.25	3.58	9.34	**0.39**
	Params (k)	282.10	239.94	237.33	349.31	523.95	1077.87	214.93
	FLOPs (G)	60916.87	41801.16	24177.64	69600.96	75848.90	32400.68	**132.24**
MUUFL	Train time (s)	2388.48	311.54	2115.96	508.32	458.05	1475.47	**33.43**
	Test time (s)	59.91	4.95	41.53	27.45	9.65	25.28	**0.08**
	Params (k)	154.80	150.17	129.95	344.09	339.95	1000.69	208.84
	FLOPs (G)	4992.54	4458.64	805.43	7373.44	5051.15	3482.27	**13.43**
Trento	Train time (s)	1098.02	342.43	1457.52	133.83	390.76	1093.64	**39.17**
	Test time (s)	25.65	8.83	9.00	12.04	7.85	22.42	**0.07**
	Params (k)	216.11	149.79	128.77	343.25	336.74	902.90	207.42
	FLOPs (G)	7689.35	6206.45	1075.16	10255.69	6969.11	4836.50	**18.56**

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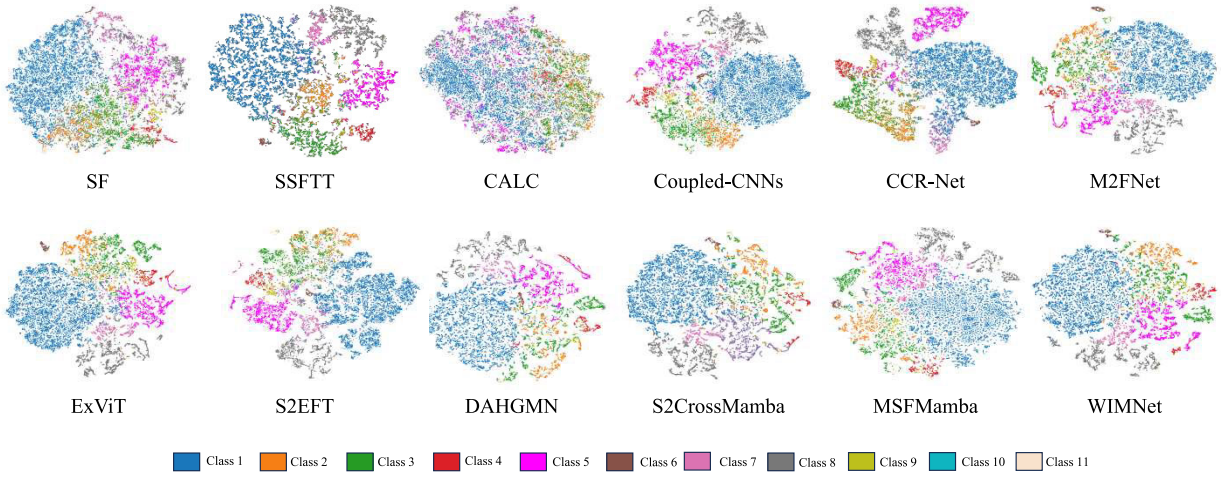


Fig. 10. t-SNE visualization on the MUUFL dataset.

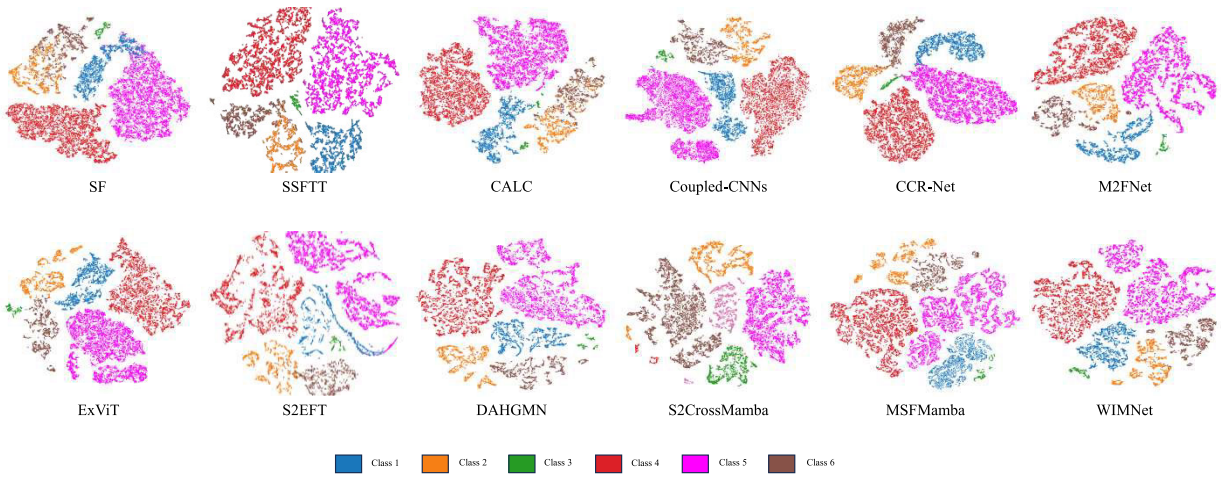


Fig. 11. t-SNE visualization on the Trento dataset.

consistent with the ground truth. The t-distributed stochastic neighbor embedding (t-SNE) [50] visualizations shown in Figs. 9–11 illustrate the feature distributions obtained by different methods. It is found that WIMNet produces a relatively organized embedding space, characterized by intraclass compactness and interclass spacing. The above results suggest that our WIMNet can effectively leverage the advantages of multimodal data and improve the representation capability of the fused features.

D. Analysis of Computational Cost and Runtime

To analyze the computational efficiency, we compare the proposed WIMNet with several transformer-based and Mamba-based multimodal methods in terms of training time, testing time, number of parameters, and computational complexity, as shown in Table VI. Since our WIMNet adopts whole-image inputs, the FLOPs are computed on entire images to ensure a fair and realistic comparison of the overall computational load. For FLOPs evaluation, we compute the FLOPs of a single patch for patch-based baselines and then multiply it by the total number of patches required to cover the entire image during inference for patch-based baselines. For runtime evaluation, the training and testing times are measured on the complete datasets under the same hardware environment.

In the Houston2013 dataset, WIMNet achieves the highest computational efficiency among all compared methods. Compared with S2CrossMamba, which attains the shortest training time (234.60 s) and testing time (3.58 s) among the compared methods, our WIMNet further reduces the training time to 201.27 s (a 14.2% reduction) and the testing time to 0.39 s (an 89.1% reduction). Regarding computational complexity, WIMNet records only 132.24 G FLOPs, which accounts for merely 0.55% of S2EFT (24 177.64 G), which attains the lowest FLOPs among the compared methods. In the MUUFL dataset, WIMNet achieves remarkable computational efficiency. Our WIMNet achieves a training time of 33.43 s and a testing time of 0.08 s, representing the shortest runtime among all compared methods. Compared with ExViT, which achieves the best performance among the compared methods with 311.54 s for training and 4.95 s for testing, WIMNet reduces the training and testing times by 89.3% and 98.4%, respectively. In terms of computational complexity, WIMNet records only 13.43 G FLOPs, which accounts for merely 1.67% of S2EFT (805.43 G), which attains the lowest FLOPs among compared

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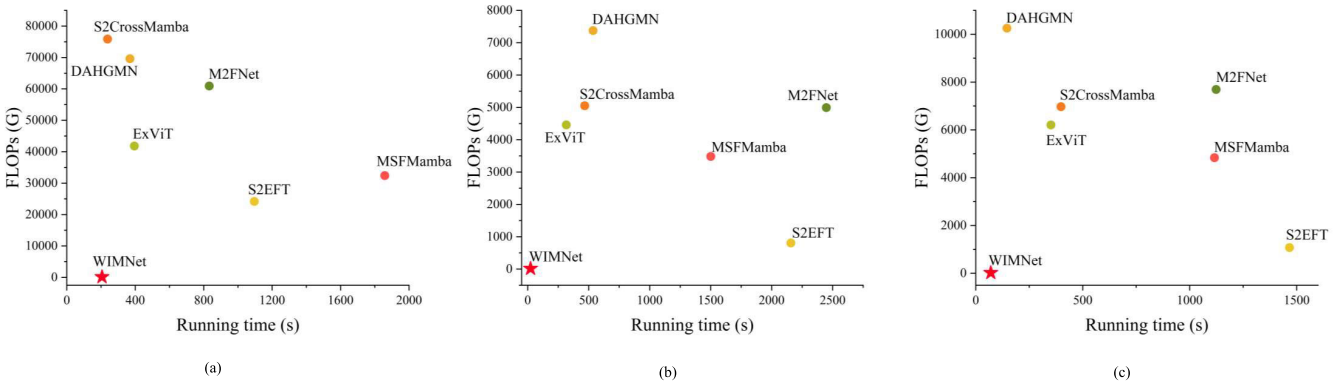


Fig. 12. Comparison of computational cost versus runtime on different methods. (a) Houston2013. (b) MUUFL. (c) Trento.

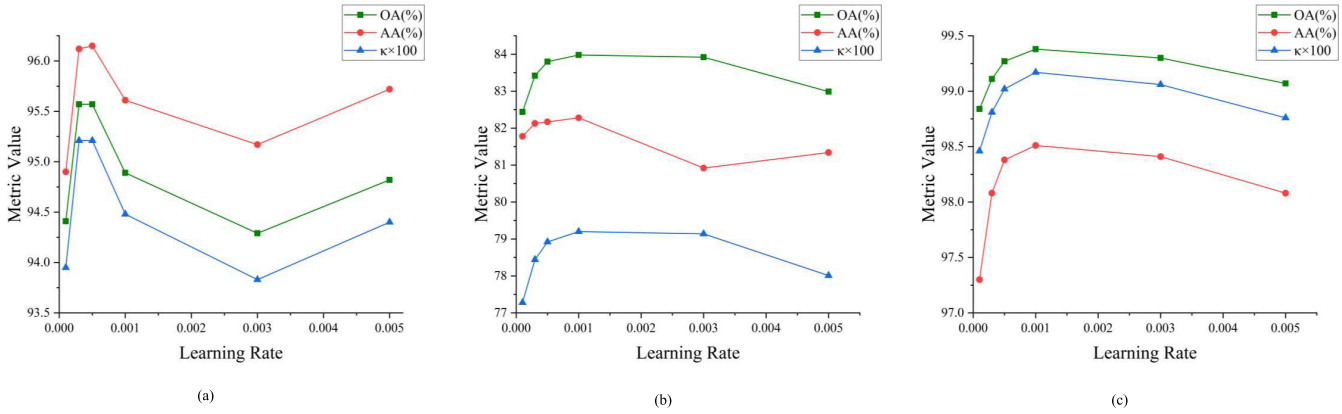


Fig. 13. Parameter sensitivity analysis of learning rate. (a) Houston2013. (b) MUUFL. (c) Trento.

methods. For the Trento dataset, our WIMNet achieves a training time of 39.17 s and a testing time of 0.07 s, representing the shortest runtime across all compared methods. Compared with DAHGMN, which achieves the shortest training time (133.83 s), and S2CrossMamba, which achieves the shortest testing time (7.85 s) among compared methods, our WIMNet reduces the training and testing times by 70.7% and 99.1%, respectively. Regarding computational complexity, WIMNet records only 18.56 G FLOPs, which accounts for merely 1.73% of S2EFT (1075.16 G), which attains the lowest FLOPs among compared methods. As shown in Fig. 12, it is evident that our WIMNet outperforms the compared methods in terms of both FLOPs and running time. Across all three datasets, WIMNet achieves the shortest training and testing times as well as the lowest FLOPs, demonstrating outstanding runtime efficiency. In addition, WIMNet exhibits well-controlled parameters, achieving a better balance between efficiency and performance.

E. Parameter Sensitivity Analysis

The learning rate plays a crucial role in determining how the model weights are updated during training. An appropriate learning rate facilitates stable convergence toward a global or local optimum. Therefore, experiments were conducted on the three datasets to determine the optimal learning rate for WIMNet. The learning rate set is $1e^{-4}$, $3e^{-4}$, $5e^{-4}$, $1e^{-3}$, $3e^{-3}$, and $5e^{-3}$ and the variations and results in OA, AA, and Kappa are illustrated in Fig. 13. For the Houston2013 dataset, the model performance improves as the learning rate increases and reaches its optimal value at $5e^{-4}$. Beyond this point, further increases in the learning rate cause a noticeable decline in performance, indicating that an excessively large learning rate leads to unstable convergence and degraded accuracy. This suggests that WIMNet benefits from a relatively moderate learning rate when trained on the Houston dataset. For the MUUFL dataset, the model performance gradually improves as the learning rate increases and tends to stabilize after $1e^{-3}$. When the learning rate exceeds this value, the AA shows a slight decline. For the Trento dataset, the model performance shows a continuous upward trend as the learning rate increases and reaches its peak at $1e^{-3}$. When the learning rate continues to increase, the model accuracy slightly decreases, indicating that an excessively high learning rate slightly affects the stability of convergence. Overall, WIMNet prefers a moderate learning rate of $5e^{-4}$ for the Houston dataset, while a higher learning rate of $1e^{-3}$ yields optimal performance on both MUUFL and Trento datasets.

In addition, we also analyzed the effect of group number, which influences the processing of HSI data by SSMB, affecting both performance and memory usage. As shown in Table VII, the model achieved the best classification accuracy

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TABLE VII
PARAMETER SENSITIVITY ANALYSIS OF GROUP NUMBER ON THREE DATASETS

Dataset	Group number	OA (%)	AA (%)	$\kappa\times100$
Houston2013	2	94.98	95.83	94.58
	4	**95.57**	**96.15**	**95.21**
	8	94.03	95.00	93.54
	16	94.58	95.48	94.14
MUUFL	2	83.92	81.25	79.01
	4	**83.98**	**82.28**	**79.20**
	8	82.33	81.00	77.03
	16	81.99	80.66	76.70
Trento	2	99.25	98.28	98.99
	4	**99.38**	**98.51**	**99.17**
	8	99.10	97.70	98.80
	16	99.32	98.32	99.09

TABLE VIII
ABLATION STUDY OF MODEL CONFIGURATIONS ON THREE DATASETS

Dataset	Configuration	OA (%)	AA (%)	$\kappa\times100$
Houston2013	Without MLCB	94.81	95.41	94.39
	Without SMM	94.69	95.60	94.26
	Without MLMB	93.77	94.59	93.27
	Full model	**95.57**	**96.15**	**95.21**
MUUFL	Without MLCB	80.35	79.04	74.68
	Without SMM	83.38	81.12	78.35
	Without MLMB	82.20	79.14	76.74
	Full model	**83.98**	**82.28**	**79.20**
Trento	Without MLCB	99.14	98.36	98.85
	Without SMM	99.09	98.49	98.78
	Without MLMB	97.58	95.92	96.78
	Full model	**99.38**	**98.51**	**99.17**

with a group number of 4 across all datasets. When set to 2, the GPU memory requirement increased without improving performance, making group number 4 the optimal choice.

F. Ablation Study

1) Component Evaluation of WIMNet: MLCB, SMM, and MLMB are the three key components in our approach. To evaluate their effectiveness, we conduct ablation experiments across the three datasets. As shown in Table VIII, the full model consistently yields the best overall performance across all three datasets, confirming the effectiveness of jointly incorporating all proposed components. When MLCB is removed, a noticeable performance degradation is observed, particularly on the MUUFL dataset. Removing SMM leads to a consistent decline in classification accuracy, suggesting that spectral-guided channel reweighting is beneficial for enhancing the spatial representation. Removing MLMB results in a notable performance drop on both Houston and Trento datasets, highlighting its crucial role in the joint classification of HSI and LiDAR. The results demonstrate that the integration of MLCB, SMM, and MLMB significantly contributes to achieving superior classification performance.

2) Input Mode Evaluation: To validate the effectiveness of the whole-image input strategy in the proposed WIMNet, we evaluate it under both whole-image and patch-based input modes in terms of classification accuracy, FLOPs, and running time. As shown in Table IX, the whole-image input consistently achieves the best classification performance on all three datasets. Compared with the best patch-based input setting, the whole-image mode improves OA by 2.37%, 8.13%, and 0.37% on Houston2013, MUUFL, and Trento, respectively. Similar gains can also be observed in terms of AA and Kappa. Notably, the whole-image input significantly reduces the computational burden. Compared with patch-based inputs, it reduces FLOPs by more than 40 times even under the smallest patch size, while also achieving much shorter training and testing time. The whole-image mode requires only 8.0%, 1.27%, and 1.19% of the inference time of the best patch-based setting on Houston2013, MUUFL, and Trento, respectively. These results validate the effectiveness of the whole-image input strategy in the proposed WIMNet.

3) SSMB Design Evaluation: To further analyze the effectiveness of the SMM, we compare it with bidirectional modeling and a pooled cross-attention variant [54] on the Trento dataset in terms of classification accuracy and GPU memory usage. Bidirectional modeling uses the same approach as SMM but in both directions, where spectral features guide spatial features while spatial features simultaneously guide spectral features. As shown in Table X, our SMM achieves the highest classification accuracy among the compared variants while maintaining the lowest computational cost, demonstrating that unidirectional SMM effectively balances accuracy and computational efficiency. t-SNE visualization, shown in Fig. 14, demonstrates that bidirectional modeling exhibits some class overlap, and Pooled Cross-Attention partially mixes classes. In contrast, our SMM produces tighter within-class clustering and clearer separation between classes.

4) Mamba Variants Analysis: We replace the standard Mamba block in WIMNet with three representative Mamba variants, including Vim [51], VSS [52], and LocalMamba [53], to evaluate their effects on classification performance and computational efficiency. The results are summarized in Table XI. WIMNet with the standard Mamba block achieves the best overall balance between classification accuracy and computational efficiency across all datasets. Although the VSS-based variant yields the shortest training and testing times, it exhibits inferior classification accuracy and higher computational cost. Meanwhile, both Vim and LocalMamba variants introduce additional complexity without classification performance gains. These outcomes suggest that the standard Mamba block is a suitable and efficient choice for WIMNet on joint hyperspectral and LiDAR classification tasks.

G. Effects of Loss Function

To validate the effectiveness of the proposed LSCE loss, a comparative experiment is conducted against the single CE loss under the same network configuration. As shown in Fig. 15, the proposed LSCE consistently achieves higher classification performance across all evaluation metrics. Compared with the standard CE loss, our LSCE demonstrates more stable optimization behavior and improved accuracy on both Houston2013 and MUUFL datasets, while maintaining comparable performance on the Trento dataset. These results indicate that incorporating label smoothing effectively mitigates model overconfidence and enhances convergence stability, leading to a more balanced and reliable training process.

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TABLE IX

EVALUATION OF WIMNET WITH DIFFERENT PATCH SIZES ON CLASSIFICATION PERFORMANCE, COMPUTATIONAL COST, AND RUNTIME

Dataset	Patch size	OA (%)	AA (%)	$\kappa\times$ 100	FLOPs (G)	Train time (s)	Test time (s)
Houston2013	7×7	92.79	94.03	92.21	6169.76	322.81	5.82
	11×11	92.25	93.55	91.63	15484.24	315.87	4.87
	15×15	93.20	94.43	92.65	29040.43	427.38	6.12
	Whole image	**95.57**	**96.15**	**95.21**	**132.24**	**201.27**	**0.39**
MUUFL	7×7	73.55	79.01	68.77	624.20	379.61	6.32
	11×11	75.85	79.03	71.35	1568.71	438.26	6.55
	15×15	73.82	70.35	68.68	2943.66	632.97	9.33
	Whole image	**83.98**	**82.28**	**79.20**	**13.43**	**33.43**	**0.08**
Trento	7×7	98.77	97.69	98.21	861.54	349.5	6.92
	11×11	99.01	98.39	98.57	2167.30	377.4	5.88
	15×15	99.01	97.25	98.56	4350.53	540.22	8.78
	Whole image	**99.38**	**98.51**	**99.17**	**18.56**	**39.17**	**0.07**

TABLE X

COMPARISON OF SSMB VARIANTS ON THE TRENTO DATASET

SSMB Type	OA (%)	AA (%)	$\kappa \times 100$	GPU Memory Usage (GB)
Bidirectional modeling	99.31	98.68	99.08	5.45
Pooled cross-attention	99.37	98.50	99.15	3.69
Ours	**99.38**	**98.51**	**99.17**	**3.58**

TABLE XI

ABLATION STUDY OF REPLACING THE STANDARD MAMBA BLOCK WITH DIFFERENT MAMBA VARIANTS ON THREE DATASETS

Dataset	Mamba Block	GFLOPS	Para (k)	Train time (s)	Test time (s)	OA (%)	AA (%)	$\kappa\times$100
Houston2013	Vim	161.15	291.09	368.03	0.55	95.49	96.21	95.13
	VSS	252.49	322.87	**161.46**	**0.31**	95.21	95.96	94.83
	Local	140.69	271.59	255.54	0.38	93.77	94.69	93.27
	Standard	**132.24**	**214.93**	201.27	0.39	**95.57**	**96.15**	**95.21**
MUUFL	Vim	16.73	284.97	51.59	0.09	82.70	80.76	77.58
	VSS	23.46	317.30	**17.91**	**0.06**	82.69	81.22	77.56
	Local	23.00	265.31	53.87	0.09	82.18	80.16	76.82
	Standard	**13.43**	**208.84**	33.43	0.08	**83.98**	**82.28**	**79.20**
Trento	Vim	22.85	279.42	55.06	0.10	99.35	98.49	99.13
	VSS	32.54	316.53	**24.74**	**0.05**	99.28	98.43	99.10
	Local	31.93	264.53	69.30	0.09	99.30	98.21	99.06
	Standard	**18.56**	**207.42**	39.17	0.07	**99.38**	**98.51**	**99.17**

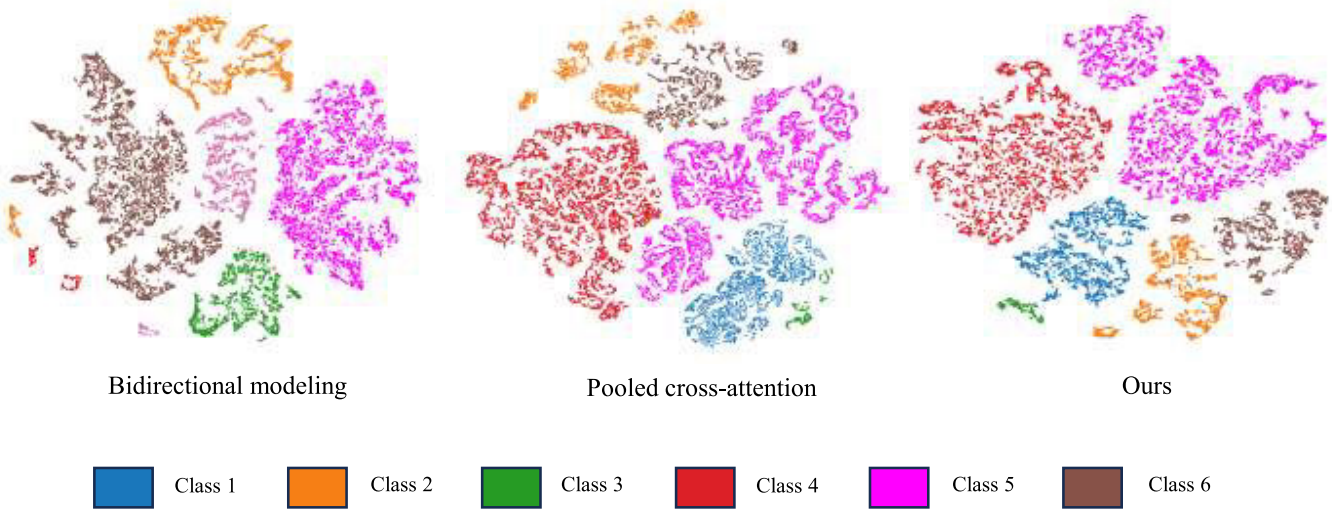


Fig. 14. t-SNE visualization of features on the Trento dataset for different SSMB types.

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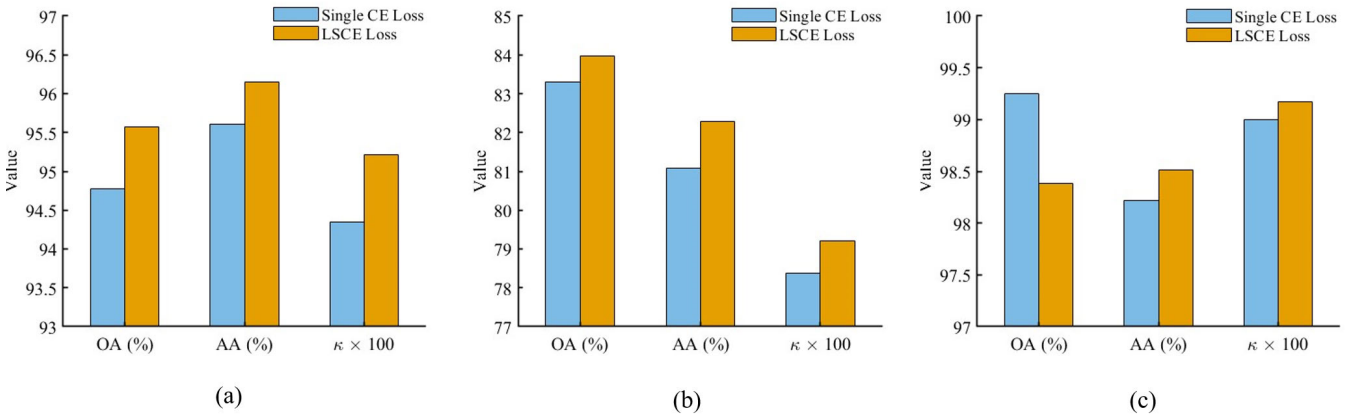


Fig. 15. Effects of loss function. (a) Houston2013. (b) MUUFL. (c) Trento.

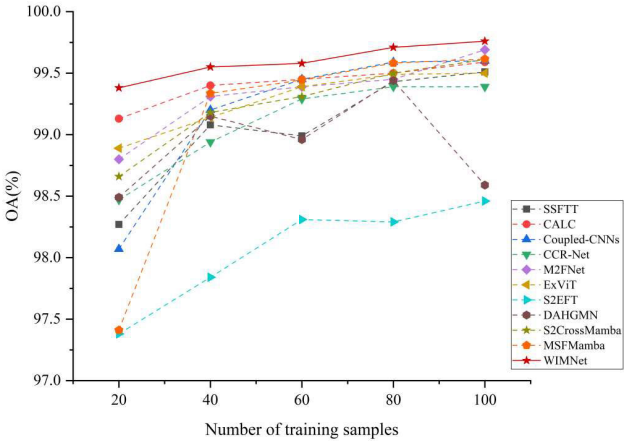


Fig. 16. Analysis of training sample sizes.

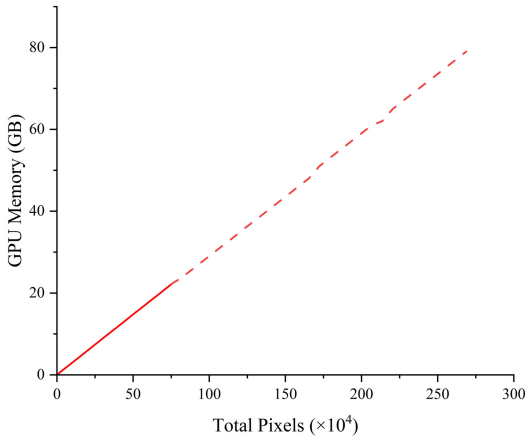


Fig. 17. Relationship between total pixels and GPU memory usage with 64 HSI channels.

H. Analysis of Training Sample Sizes

The number of training samples affects the classification performance in RS applications. To assess the performance of the proposed WIMNet, experiments are conducted using the Trento dataset with varying training sample sizes: 20, 40, 60, 80, and 100. The experimental results, shown in Fig. 16, indicate that WIMNet consistently outperforms other competitors when the number of training samples is small. Fig. 16 does not include SF due to its lower OA. As the number of samples increases, WIMNet exhibits a steady improvement in classification accuracy. The proposed model achieves high performance with fewer samples, maintaining competitive accuracy levels across different sample sizes, which highlights its effectiveness in real-world RS scenarios of limited labeled data.

V. Discussion

In this section, we discuss the practical limitations of the whole-image input strategy. While the whole-image-based framework improves inference efficiency and reduces computational redundancy, it increases GPU memory usage due to the high spectral dimensionality of hyperspectral and LiDAR data.

TABLE XII
GPU Memory Consumption Across Different Datasets

Dataset	GPU Memory Usage (GB)	
	Original	Tiling Strategy
Houston2013	20.97	9.59
MUUFL	2.69	–
Trento	3.58	–

To better analyze this limitation, we conduct a quantitative analysis of GPU memory consumption across different datasets. As shown in Table XII, whole-image training can be directly conducted with GPU memory consumption below 4 GB for relatively lower resolution datasets such as MUUFL and Trento, indicating that the proposed method is practical for moderate-scale inputs. For the higher resolution dataset Houston2013, we adopt a tiling strategy in which the input image is divided into two overlapping subregions that are processed sequentially with gradients accumulated before a single parameter update, while whole-image inference is retained at test time. This strategy reduces the GPU memory consumption from approximately 21 to below 10 GB, effectively reducing GPU memory usage. We further analyze the relationship

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between GPU memory consumption and the total number of input pixels of the proposed method without the tiling strategy. As shown in Fig. 17, the results indicate that under a 24-GB GPU, the proposed WIMNet can effectively handle high-resolution images, with memory usage growing in a predictable manner as the image size increases. This demonstrates that the proposed WIMNet remains feasible for higher resolution datasets, and the memory demand is acceptable with current GPU hardware. In the future, we hope that more techniques for optimizing training memory usage could improve scalability and address this limitation.

VI. Conclusion

In this article, we propose WIMNet for joint classification of hyperspectral and LiDAR data. By directly processing an entire image, WIMNet reduces redundant computation and improves classification efficiency. WIMNet adopts a dual-branch architecture consisting of JSMB and MLMB, which extract spectral–spatial and elevation features, respectively. These features are integrated through MFFM for the final classification. To enhance the interaction between spectral and spatial features, we design the SSMB, which introduces the SMM to achieve spectral-guided spatial modeling. Final experiments on three public benchmarks demonstrate that the proposed WIMNet outperforms 11 state-of-the-art methods in classification accuracy. In terms of computational efficiency, WIMNet achieves the shortest running time and the lowest FLOPs across all datasets. On the MUUFL dataset, the training and testing times are reduced by 89.3% and 98.4%, respectively, and the FLOPs are only 1.67% compared with S2EFT. We believe that the whole-image-based strategy with Mamba architectures holds great potential in the field of multimodal RS, and we will explore their applications to other RS tasks in future work.

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